

Lists and indexing

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace an update for Lists and indexing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Lists and indexing. Proposed explanation: “Length is the last nonnegative index”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Distinguish a value from an index

Predict the three outputs. If the second print uses values[3] instead, where does execution stop and which later output is not produced?

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Separate index and value

For counts = [4, 7, 4], give the length, valid nonnegative indices and value at index 1. Explain how the two items with value 4 differ.

My reasoning and evidence

6. W2 Trace an update

Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting list, its length and the value at index 2. Predict before running.

My reasoning and evidence

7. W3 Repair a position mistake

A learner loops with for count in counts and tries to read counts[count] for counts = [4, 7, 4]. Identify the first attempted index and error. Explain how to print each value directly.

My reasoning and evidence

8. W4 Process new data independently

Start with readings = [3, 0, 3, 2]. Replace the second item with 5 using its index. Write a loop that adds all updated readings and prints one final total. Trace every update of the total. Then test the summing part alone on [] and [6].

My reasoning and evidence

9. W5 Keep the information you need

Compare [1, 5, 2] and [5, 1, 2]. State their totals and first items. Would storing only the total answer which first item was larger? Explain.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled lists and indexing evidence sheet to Python readiness portfolio. Show how you can distinguish a list index from the value stored at that position. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Lists and indexing

Indexes are positions, not values. A list of shelf counts is read by position.

Python 3: `counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).`

Zero-based list positions

	Index 0	Index 1	Index 2
Stored counts	6	1	8

Read a cell using both its row and column labels.

Lists and indexing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw positions

Indexes 0,1,2 contain 6,1,8 respectively.



2. Read two positions

Index 0 returns 6; index -1 returns the final value 8.



3. Check length boundary

Length is 3, but index 3 is beyond the final valid positive index 2.

CHANGE ONE THING

Replace the second print with `print(counts[3])`.

Explain the changed result using the matching lesson.

Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.